Financial Calculations Reference

1. EMI Ratio (FOIR — Fixed Obligation to Income Ratio)

EMI Ratio (%) = (Existing Loan EMI ÷ Monthly Income) × 100

Component	Description	Example	Result
Existing Loan EMI	Fixed monthly loan repayment	₹ 12,000	—
Monthly Income	Gross monthly income	₹ 85,000	—
EMI Ratio	% of income on loans	12000 ÷ 85000 × 100	14.12%

EMI Ratio Range	Classification	Loan Eligibility (RBI Standard)
Below 30%	Excellent	High eligibility — qualifies easily
30% – 40%	Good	Moderate eligibility — qualifies with conditions
40% – 50%	Caution	Low eligibility — banks may hesitate
Above 50%	Critical	Not eligible — debt burden too high

2. Monthly Savings Rate

Savings Rate (%) = (Monthly Income − Monthly Expense) ÷ Monthly Income × 100

Component	Description	Example	Result
Monthly Income	Gross monthly income	₹ 85,000	—
Monthly Expense	Total monthly spending	₹ 45,000	—
Monthly Surplus	Income minus expense	₹ 40,000	—
Savings Rate	% of income saved	40000 ÷ 85000 × 100	47.06%

Savings Rate	Classification	Recommendation
Above 20%	Excellent	Strong savings habit — focus on investments
10% – 20%	Good	Healthy — room for improvement
0% – 10%	Poor	Low savings — reduce discretionary spending
Below 0%	Critical	Deficit — expenses exceed income

3. Credit Utilization Ratio (optional — 0% if no credit card)

Credit Utilization (%) = (Credit Card Used ÷ Credit Card Limit) × 100

Component	Description	Example	Result
Credit Card Used	Outstanding balance	₹ 60,000	—
Credit Card Limit	Sanctioned credit limit	₹ 2,00,000	—
Credit Utilization	% of limit used	60000 ÷ 200000 × 100	30%

Utilization Range	CIBIL Impact	Recommendation
Below 30%	Positive — improves score	Ideal — maintain this
30% – 60%	Neutral — no major impact	Acceptable — try to reduce
Above 60%	Negative — lowers score	High risk — pay down balance

4. Projected Annual Savings

Projected Annual Savings (₹) = (Monthly Income − Monthly Expense) × 12

Component	Description	Example	Result
Monthly Income	Gross monthly income	₹ 85,000	—
Monthly Expense	Total monthly spending	₹ 45,000	—
Monthly Surplus	Net cash per month	₹ 40,000	—
Projected Annual Savings	Surplus × 12 months	40000 × 12	₹ 4,80,000

5. Monthly Surplus & Deficit Flag

Monthly Surplus (₹) = Monthly Income − Monthly Expense

is_deficit = True if Monthly Surplus < 0 else False

Condition	is_deficit	AI Behavior
Income > Expense	False	Normal report — investments and loan advice given
Income = Expense	False	Savings rate = 0% — caution advised

Income < Expense	True	CRITICAL — no loans, no investments, stabilise budget first

6. 20 / 4 / 10 Car Loan Rule (triggered during follow-up chat only)

Rule 1 — Down Payment % = (Down Payment ÷ Car Price) × 100 ≥ 20%

Rule 2 — Tenure Check : Loan Tenure in years ≤ 4 years

Rule 3 — Car Expense % = (Monthly Car EMI ÷ Monthly Income) × 100 ≤ 10%

Overall Pass = Rule 1 PASS AND Rule 2 PASS AND Rule 3 PASS

Rule	Formula	Threshold
20 — Down Payment	Down Payment ÷ Car Price × 100	Must be ≥ 20%
4 — Tenure	Loan Tenure in years	Must be ≤ 4 yrs
10 — Car Expense	Car EMI ÷ Monthly Income × 100	Must be ≤ 10%
Overall	All 3 rules must pass	PASS or FAIL